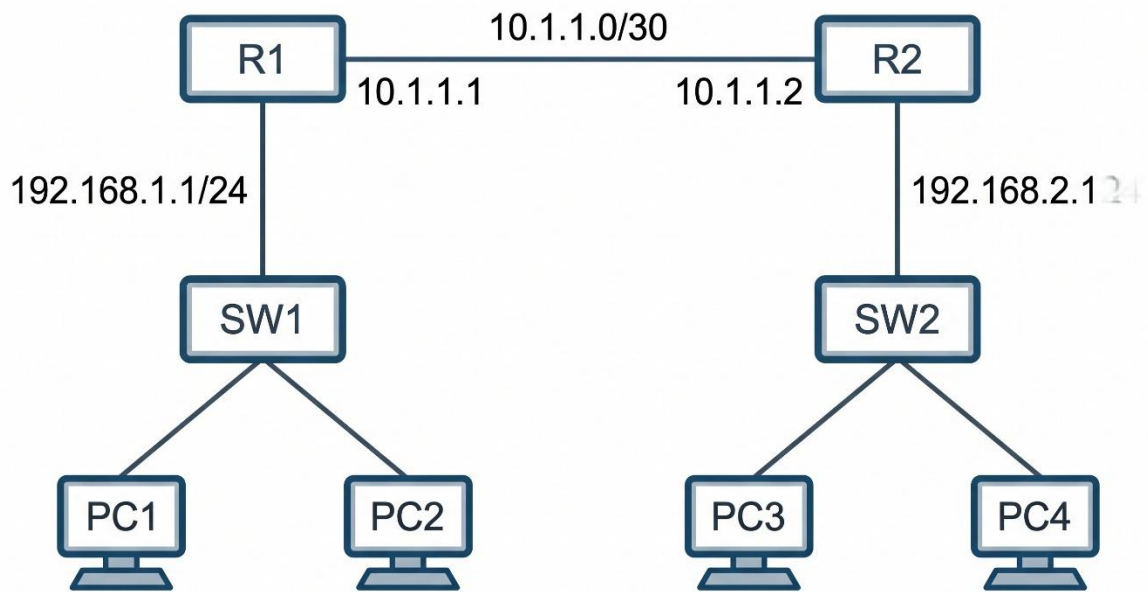




## Static Routing + SSH

### Ip Address plan

| Device | Interface | IP Address   | Mask | Gateway     |
|--------|-----------|--------------|------|-------------|
| R1     | LAN       | 192.168.1.1  | /24  | —           |
| R1     | R1-R2     | 10.1.1.1     | /30  | —           |
| R2     | R2-R1     | 10.1.1.2     | /30  | —           |
| R2     | LAN       | 192.168.2.1  | /24  | —           |
| SW1    | VLAN 1    | 192.168.1.10 | /24  | 192.168.1.1 |
| SW2    | VLAN 1    | 192.168.2.10 | /24  | 192.168.2.1 |

## PC IP Address

| PC  | IP Address   | Mask          | Gateway     |
|-----|--------------|---------------|-------------|
| PC1 | 192.168.1.11 | 255.255.255.0 | 192.168.1.1 |
| PC2 | 192.168.1.12 | 255.255.255.0 | 192.168.1.1 |
| PC3 | 192.168.2.11 | 255.255.255.0 | 192.168.2.1 |
| PC4 | 192.168.2.12 | 255.255.255.0 | 192.168.2.1 |

### Note:

#### Transport input SSH(Command )

The transport input ssh command configures a network device's Virtual Terminal (VTY) lines to only accept Secure Shell (SSH) connections for remote management.

#### ip ssh version 2(Command)

For specific version of ssh

#### ip default-gateway 192.168.1.1(Command)

Arko network ma jana

#### crypto key generate rsa modulus 2048(command)

An RSA key size of 2048 bits is the industry standard because it provides the optimal balance between high security and fast processing performance.

#### description Management IP(Command under interface)

The description command under an interface is used to label and document the purpose of the interface, while the ip address command assigns the actual network identity.

## R1 Configuration

enable

configure terminal

R1(config)#hostname R1

LAN interface

R1(config)#interface gigabitEthernet 0/0

description LAN-interface

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

R1(config)#interface gigabitEthernet 0/1

description R1-R2

ip address 10.1.1.1 255.255.255.252

no shutdown

exit

Static route to R2 LAN

R1(config)#ip route 192.168.2.0 255.255.255.0 10.1.1.2

### SSH configuration

R1(config)#ip domain-name ccna.local

R1(config)#username admin privilege 15 secret cisco

```
R1(config)#crypto key generate rsa modulus 2048
```

```
R1(config)#ip ssh version 2
```

```
R1(config)#line vty 0 15
```

```
login local
```

```
transport input ssh
```

```
exit
```

```
end
```

```
write memory
```

## **R2 Configuration**

```
enable
```

```
configure terminal
```

```
R2(config)#hostname R2
```

```
R2(config)#interface gigabitEthernet 0/0
```

```
description LAN-interface
```

```
ip address 192.168.2.1 255.255.255.0
```

```
no shutdown
```

```
exit
```

```
R2(config)#interface gigabitEthernet 0/1
```

```
description R2 to R1
```

```
ip address 10.1.1.2 255.255.255.252
```

```
no shutdown
```

```
exit
```

Static route to R1 LAN

```
R2(config)#ip route 192.168.1.0 255.255.255.0 10.1.1.1
```

SSH configuration

```
R2(config)#ip domain-name ccna.local
```

```
R2(config)#username admin privilege 15 secret cisco
```

```
R2(config)#crypto key generate rsa modulus 2048
```

```
R2(config)#ip ssh version 2
```

```
R2(config)#line vty 0 15
```

```
login local
```

```
transport input ssh
```

```
exit
```

```
end
```

write memory

## **SW1 Configuration**

enable

configure terminal

Sw1(config)#hostname SW1

Sw1(config)#interface vlan 1

Sw1(config-if)#description Management IP

Sw1(config-if)# ip address 192.168.1.10 255.255.255.0

Sw1(config-if)#no shutdown

exit

Sw1(config)#ip default-gateway 192.168.1.1

### **SSH configuration**

Sw1(config)#ip domain-name ccna.local

Sw1(config)#username admin privilege 15 secret cisco

Sw1(config)#crypto key generate rsa modulus 2048

Sw1(config)#ip ssh version 2

Sw1(config)#line vty 0 15

login local

transport input ssh

exit

end

write memory

## **SW2 Configuration**

enable

configure terminal

hostname SW2

Sw2(config)#interface vlan 1

Sw2(config-if)#description Management IP

Sw2(config-if)# ip address 192.168.2.10 255.255.255.0

Sw2(config-if)# no shutdown

exit

Sw2(config)#ip default-gateway 192.168.2.1

Sw2(config)#ip domain-name ccna.local

```
Sw2(config)#username admin privilege 15 secret cisco
```

```
Sw2(config)#crypto key generate rsa modulus 2048
```

```
Sw2(config)#ip ssh version 2
```

```
Sw2(config)#line vty 0 15
```

```
login local
```

```
transport input ssh
```

```
exit
```

```
end
```

```
write memory
```

## **T-shoot command**

### **R1**

```
show ip interface brief
```

```
show ip route
```

```
show running-config
```

```
show ip ssh
```

```
show users
```

### **R2**

```
show ip interface brief
```

show ip route

show ip ssh

## **SW1/SW2**

show ip interface brief

show vlan brief

show ip ssh

show running-config